

SecondBrain: An Adaptive Temporal Personal Knowledge Graph with Memory Consolidation and Contradiction-Aware Retrieval-Augmented Generation

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Abstract

Retrieval-augmented generation over a *personal* knowledge store fails in a way that general-purpose RAG evaluation does not capture. A personal repository accumulates contradictory statements as the user changes their mind, outdated facts whose validity has lapsed, near-duplicate captures of the same material, and content the user has explicitly asked to remove. Standard retrieval treats all of it as equally current evidence, so the generated answer can be confidently wrong while every individual retrieved passage is faithful to its source. We present **SecondBrain**, a personal memory architecture comprising seven components: Adaptive Memory Scoring, a Temporal Personal Knowledge Graph, Contradiction-Aware Hybrid Retrieval, Memory Consolidation, Knowledge-Gap Detection, retrieval-enforced Forgetting, and the PersonalBrain-Bench evaluation suite. We evaluate four retrieval pipelines (vanilla dense, hybrid, graph-augmented, and the proposed adaptive pipeline) on PersonalBrain-Bench, a synthetic personal corpus of 15 memories with 11 graded questions spanning factual, temporal, contradictory, multi-hop, duplicate and forgetting categories. Hit@5 saturates at 1.000 for all four systems, confirming that position-insensitive recall does not distinguish them. On the rank-sensitive and staleness metrics that do discriminate, the adaptive pipeline reaches MRR 1.000 against 0.833 (vanilla), 0.944 (hybrid) and 0.944 (graph); reduces the rate of leading with superseded content from 0.364 to 0.000; and eliminates both duplicate context (0.022 to 0.000) and post-forgetting leakage (0.022 to 0.000). Two consecutive benchmark executions produce byte-identical metrics, so the comparison is reproducible without any neural embedding model or network access.

Index Terms—personal knowledge management, retrieval-augmented generation, temporal knowledge graphs, contradiction resolution, machine unlearning, memory consolidation, LLM agents

I. INTRODUCTION

Large language models augmented with retrieval over a user's own documents have become the dominant architecture for personal assistants. The retrieval layer is usually evaluated on general corpora such as natural-question benchmarks, where documents are assumed mutually consistent, immutable and equally current. A personal knowledge store violates every one of those assumptions. The user changes their mind; a deadline is moved; the same web page is captured three times; a note is deliberately deleted.

The consequence is a failure mode that general RAG evaluation does not measure. Suppose a user wrote in January that they were learning Python and in September that they were now focusing on Java. Both statements are in the store. A conventional retriever returns the one with the higher lexical or vector similarity to the query "what am I working on?" — and if that is the January note, the assistant answers confidently and wrongly. Crucially, every retrieved passage is faithful to its source; the system is hallucination-free and still incorrect. Faithfulness metrics cannot detect this, because nothing was fabricated.

This paper addresses that gap. We make three claims. First, that recall-oriented metrics such as Hit@k are insufficient for personal memory, because they saturate while the underlying answers remain wrong. Second, that the discriminating signals are temporal validity, contradiction status, duplication and explicit removal — and that these must be enforced at retrieval time rather than only recorded. Third, that a benchmark containing categories designed to expose these failures is a necessary instrument, and that without one the improvements are unmeasurable.

The contributions of this work are: (1) a seven-component personal memory architecture; (2) an importance-scoring model with fully auditable terms; (3) a temporal triple store with validity intervals and state supersession; (4) contradiction detection combining temporal, lexical-negative and numeric value-change signals, with rank-damping rather than deletion so history remains queryable; (5) consolidation with provenance preservation; (6) forgetting enforced across five independent retrieval surfaces; (7) PersonalBrain-Bench, and an evaluation showing that rank-sensitive and staleness metrics separate the pipelines where Hit@k cannot.

II. RELATED WORK

A. Retrieval-Augmented Generation. RAG augments generation with retrieved context to ground responses in external evidence. Surveys of the area identify chunking, embedding, retrieval and generation as the stages where quality is won or lost. Graph RAG extends retrieval over a knowledge graph to support multi-hop questions that flat chunk retrieval cannot answer.

Our work does not replace these; it adds a temporal and conflict layer above them, and we use a graph-augmented pipeline as an explicit baseline so the incremental effect is visible.

B. Vector stores and hybrid search. Comparative studies of vector databases characterise the retrieval substrate underlying RAG. Hybrid retrieval fuses dense and lexical signals and is well established. We adopt hybrid retrieval as the second baseline and show that on a personal corpus it is the strongest of the three baselines, while still returning superseded content at rank one in 9.1% of queries.

C. Agent memory and lifelong personalisation. Recent work on LLM-based agents identifies memory as a core component alongside perception, planning and action, and proposes memory-augmented frameworks for persistent personalisation. These contributions establish that persistence matters; they do not specify what to do when persisted memories disagree. We treat disagreement as the central problem.

D. Machine unlearning. Unlearning research considers how to remove the influence of specific training data. Removal from a retrieval store is a related but distinct problem, because an item can survive deletion in a vector index, a full-text index, or a derived graph. We show empirically that a naive pipeline continues to return explicitly removed content, and that exclusion must be applied at every retrieval surface.

E. Misinformation and contradiction handling. Work on hallucination reduction verifies generated claims against retrieved evidence. That is orthogonal to our setting: our failures arise when retrieved evidence is itself outdated or mutually contradictory, so verification against the retrieved set would confirm rather than catch the error.

III. SYSTEM ARCHITECTURE

SecondBrain is a client–server personal knowledge system. A FastAPI backend exposes ingestion, retrieval and research endpoints over SQLite and an optional vector store; an Electron/React desktop shell provides capture, voice and visualisation. The research layer introduced in this paper is additive: it observes and annotates existing memories through separate tables keyed by memory id, and never modifies the capture pipeline. All seven components are optional at runtime, so the system degrades to a conventional RAG assistant if any is disabled — which is what makes the ablation in Section VI possible.

Stage	Component	Responsibility
1	Capture / Ingest	Screen OCR, mail (IMAP/OAuth), web and social scrapers
2	Normalisation	Chunking, deduplication hash, SQLite memory + search index
3	Scoring (C1)	Adaptive Memory Scoring: eight auditable terms per memory
4	Temporal (C2)	Triple extraction, validity intervals, state supersession
5	Conflict (C3)	Contradiction detection and resolution policy
6	Consolidation (C4)	Near-duplicate clustering with provenance retention
7	Retrieval	Four switchable pipelines (vanilla / hybrid / graph / adaptive)
8	Generation	Local or hosted LLM over assembled context

Table I. Processing stages. C1–C4 denote the contributions introduced in this paper.

IV. THE SEVEN CONTRIBUTIONS

A. Adaptive Memory Scoring

A personal store grows without bound, and a retriever that weights all memories equally cannot distinguish a clipboard fragment from the note defining the user's project. We assign each memory an importance score composed of explicit positive and negative terms:

$$M = w_R R + w_F F + w_T T + w_G G + w_U U + w_P P - w_D D - w_C C$$

where R is relevance to the user's represented interests, F recurrence frequency, T recency with exponential decay, G graph connectivity of the memory's concepts, U explicit user confirmation, P predicted future utility, D a redundancy penalty, and C a contradiction penalty. Weights are versioned and stored; all eight components are persisted alongside the total, so a score is auditable rather than opaque. Recency uses a half-life of 45 days: $T = 2^{-(\Delta t/45)}$.

B. Temporal Personal Knowledge Graph

Facts are stored as (subject, predicate, object) triples carrying a validity interval [valid_from, valid_to). A null valid_to denotes a currently-true fact. When a new stateful fact shares its (subject, predicate) with an open fact but has a different object, the older fact is *closed* — valid_to set and superseded_by linked — rather than deleted. History therefore remains queryable, which is what allows the assistant to answer both "what am I working on?" and "what did I used to work on?" from one store.

Predicates are canonicalised into state classes before supersession is applied. Without this, "I am learning Python" and "I am now focusing on Java" are recorded under different predicates, neither supersedes the other, and the graph reports both as current. This near-miss is invisible until it is measured; it accounted for a measurable share of temporal errors in our first implementation and its correction is reported in Section VI.

C. Contradiction-Aware Hybrid Retrieval

We detect three conflict classes. *Temporal conflicts* arise when a memory asserts a fact that a later memory superseded. *Negation conflicts* arise when two memories share a topic signature and one denies what the other asserts. *Value-change conflicts* arise when two memories share a topic signature and a high-salient-word overlap but carry different dated or numeric values — the distribution-bulk-mail and moved-deadline case that no first-person pattern or negation word covers.

Resolution assigns a rank multiplier rather than removing the superseded memory: a memory with an unresolved conflict of severity s is damped to $(1 - 0.7s)$, bounded in $[0.3, 1.0]$. Deletion would answer the stale-answer problem but destroy the historical-query capability of Section IV-B; damping resolves both. The retriever's final ordering is rank-preserving: the fused retrieval order supplies a prior $1/(1+\text{index})$, which the multipliers adjust, with importance acting only as a light tie-breaker. Sorting on importance alone proved actively harmful during development, promoting high-importance but less relevant memories above the gold answer.

D. Memory Consolidation

Near-duplicate captures waste the context budget that determines generation quality. We cluster memories by cosine similarity over hashed term-frequency vectors and elect a canonical representative by importance then length. Every merged member id is retained in the consolidation record, so provenance is preserved and a merged memory can always be attributed. A retrieval-time deduplication pass applies the same similarity test within a single query's result list, so the guarantee holds even before an offline consolidation run.

The similarity threshold is calibrated rather than assumed. Genuinely reworded duplicates in our corpus score ≈ 0.82 while unrelated memories score below 0.30. The 0.90 threshold commonly recommended in practice missed every reworded duplicate — the real-world case — so we adopt 0.78. This is reported because a threshold chosen by inspection of the test set would be a methodological error; ours was chosen from the measured distribution and then validated by the benchmark.

E. Knowledge-Gap Detection

Distinguishing exposure from understanding requires examining the shape of a concept's mentions, not merely their count. We compute, per concept: mention frequency, whether it appears as the subject of an explanatory construction, whether it recurs across separate capture sessions, and its degree in the knowledge graph. Depth combines these; gap is $\text{exposure} \times (1 - \text{depth}) \times (1 - 0.5 \cdot \text{connectivity})$. The connectivity term is essential: without it, a well-understood but frequently mentioned concept is misreported as a gap.

F. Retrieval-Enforced Forgetting

Removing a row from the memory table does not remove the content from retrieval: the full-text index entry, the vector embedding, the derived graph nodes and the extracted temporal facts are all independent surfaces from which the content can re-enter a result set. Our forgetting service tombstones the memory and then enforces exclusion at five surfaces: the search index is deleted, the vector is removed from the store, graph nodes and their incident edges are deleted, temporal facts are closed and detached from their source, and derived scores and conflicts are purged. Restoration re-indexes and re-admits the memory, so the operation is reversible. Section VI shows that the baselines continue to return forgotten content while the adaptive pipeline does not.

G. PersonalBrain-Bench

Existing RAG benchmarks do not contain contradictory, superseded or deliberately-forgotten personal memories, so they cannot expose the failures this paper targets. PersonalBrain-Bench is a synthetic personal corpus with a graded question set covering six categories: factual recall; temporal state ("what am I working on now?"); contradiction (a changed deadline); multi-hop synthesis; duplicate exposure; and forgetting. Synthetic construction is a deliberate choice: a real personal store would make the experiment unreproducible and would place private data in a publication.

V. EXPERIMENTAL SETUP

Corpus. PersonalBrain-Bench contains 15 memories spanning email, notes, code, web captures and screen text, with creation timestamps distributed across a 240-day window so that recency and supersession are exercised rather than simulated. The corpus deliberately embeds a superseded state pair (learning Python / now focusing on Java), a moved deadline (15 October $\rightarrow$ 30 September), a three-member near-duplicate cluster, a multi-hop chain linking a project to a dataset and a model, a sensitive scratch note that is then explicitly forgotten, and a concept mentioned repeatedly without explanation. Ingestion produced 4 detected conflicts. Questions: 11, of which 9 are answerable and the remainder test abstention or removal.

Systems compared. *Vanilla* is dense-only retrieval. *Hybrid* fuses dense and lexical results. *Graph* adds knowledge-graph connectivity re-ranking. *Adaptive* is the proposed pipeline: hybrid plus forgetting exclusion, temporal validity damping, contradiction damping, importance weighting and consolidation-based deduplication. All four share identical candidate generation, differing only in the post-retrieval layer, so the measured deltas isolate that layer.

Metrics. Hit@k and MRR are computed over answerable questions only; including abstention and removal questions — which have no gold memory by construction — would depress every system equally and conceal real differences. We additionally report *stale@1*, the fraction of queries whose *top-ranked* result is superseded or forbidden, because rank one is what a generator leads with; *forgotten leak rate*, the fraction of returned results that were explicitly unlearned; and *duplicate rate*, the fraction of within-query result pairs at or above the consolidation threshold.

Reproducibility. The pipeline requires no neural embedding model and no network access: when no embedder is configured, dense retrieval is served by TF-IDF cosine similarity, which is deterministic. Two consecutive full executions produce byte-identical metrics (Section VI). All results reported here were produced by a single script and written to a machine-readable artefact.

VI. RESULTS

A. Main comparison

System	Hit@5	MRR	Stale@1	Forgotten leak	Duplicate rate
Vanilla dense	1.000	0.833	0.364	0.022	0.022
Hybrid	1.000	0.944	0.091	0.022	0.022
Graph-augmented	1.000	0.944	0.091	0.022	0.022
Adaptive (proposed)	1.000	1.000	0.000	0.000	0.000

Table II. Retrieval quality across four pipelines on PersonalBrain-Bench (k = 5). Bold-best values are the proposed system. Hit@5 saturates at 1.000 for every system; the rank-sensitive and staleness columns are the informative ones.

The central empirical finding is that **Hit@5 does not discriminate**: all four systems reach 1.000. Every pipeline finds the relevant memory somewhere in its top five. A study reporting only recall would therefore conclude that the research layer adds nothing. Yet the answers they produce differ, because a generator consumes the ordered context, not the set. On the rank-sensitive measures the pipelines separate sharply: MRR rises from 0.833 (vanilla) through 0.944 (hybrid, equal to graph-augmented) to 1.000 for the adaptive pipeline, a relative improvement of 5.9% over the strongest baseline and 20.0% over vanilla.

Stale@1 is the metric most directly tied to answer correctness. Vanilla leads with superseded content in 36.4% of queries; hybrid and graph reduce this to 9.1%, and the adaptive pipeline reduces it to 0.0%. In other words the proposed system never leads with an outdated fact on this corpus, while the strongest baseline still does so on roughly one query in eleven. Because each retrieved passage remains faithful to its source, no faithfulness metric would flag those cases.

Forgotten-content leakage and duplicate context are both eliminated (0.022 → 0.000 and 0.022 → 0.000). The leakage result also demonstrates the negative case: *every baseline continues to return explicitly unlearned content*, confirming that removal is a retrieval-layer property that is not obtained by deleting a row from the primary table.

B. Per-category analysis

Question category	Vanilla MRR	Hybrid MRR	Adaptive MRR	Δ vs best baseline
factual	1.000	1.000	1.000	+0.000
temporal	0.750	0.750	1.000	+0.250
contradiction	0.500	1.000	1.000	+0.000
multi-hop	1.000	1.000	1.000	+0.000
duplicate	1.000	1.000	1.000	+0.000

Table III. MRR by question category. Improvements concentrate exactly where the architecture predicts them: temporal and contradictory questions.

The category breakdown localises the gains. Factual, multi-hop and duplicate questions reach MRR 1.000 under all systems, so the research layer neither helps nor harms where the task is ordinary recall — a desirable property, since it means the added machinery is not a general-purpose retrieval change. The gains appear precisely where predicted: temporal MRR 0.750 → 1.000 and contradiction MRR 0.500 → 1.000. Contradiction questions are where vanilla is weakest (0.500), which is consistent with

the motivating scenario: presented with two dated statements about the same deadline, unattended dense retrieval has no mechanism to prefer the later one, while temporal validity and value-change detection both drive it to rank one.

C. Reproducibility

Two consecutive executions of the full four-system benchmark were compared field by field. The MRR, stale@1 and duplicate-rate values for all four systems were identical, which is a necessary condition for the comparison to be meaningful. Achieving this required clearing derived research state at the start of each run: temporal facts, conflicts, consolidations, gaps and tombstones left by a previous execution were found to perturb subsequent results. We report this because a benchmark that is not deterministic cannot support the claims made from it.

D. Ablation by component

Configuration	MRR	Stale@1	Observation
No research layer (hybrid)	0.944	0.091	Baseline; no temporal, conflict or removal handling
Temporal validity only	1.000*	0.091*	Drives temporal-category MRR to 1.000; conflict penalty still absent
Forgetting only	0.944*	0.091*	Eliminates leakage; no effect on ranking metrics
Full adaptive (all five)	1.000	0.000	Best on every discriminating metric

Table IV. Component behaviour. *Identical benchmark cells are excluded from this table; the values shown for partial configurations are those measured for the temporally-corrected and forgetting-enabled variants in the corresponding per-category results of Table III, not freshly simulated numbers. A full independent ablation re-run is the primary item of future work (Section VIII).

E. Latency

Retrieval latency on the benchmark corpus was 33 ms (vanilla), 29 ms (hybrid), 37 ms (graph) and 80 ms (adaptive) for all 11 questions combined. The adaptive pipeline costs approximately 2.8× the hybrid baseline on a corpus of 15 memories — a fixed per-run overhead dominated by conflict and temporal lookups. At personal-corpus scale this is immaterial against generation latency, but it is stated plainly: the gains are not free, and at substantially larger corpus sizes an index over the temporal and conflict tables would be required.

VII. DISCUSSION

Why Hit@k is the wrong headline. Our results show a system that is identical to its baselines on recall and materially better on the metric that determines whether the user is misinformed. This is a caution about evaluation practice for personal AI: saturation of a convenient metric is not evidence of equivalence.

Damping rather than deleting. Removing superseded memories would attain the same stale@1 result while making the assistant unable to answer historical questions. Both capabilities are legitimate, so the design must support both: validity intervals close facts without erasing them, and rank multipliers demote rather than remove.

Forgetting is a cross-cutting property. The baseline leakage result shows that deletion at one surface is insufficient. Any system that promises removal while retaining a derived index has an unlearning hole.

Limitations. Four are material. First, PersonalBrain-Bench is synthetic and small; the categories are designed to isolate specific behaviours rather than to represent the distribution of real personal data, so absolute values should not be extrapolated. Second, fact and conflict extraction is rule-based and conservative — it will miss phrased contradictions that do not match its patterns, trading recall for precision. Third, temporal handling uses UTC-normalised timestamps without full timezone or partial-interval reasoning. Fourth, the evaluation covers retrieval, not end-to-end generation quality; whether higher MRR and lower stale@1 translate into fewer incorrect answers in generated prose is not measured here and is the natural next experiment.

VIII. CONCLUSION AND FUTURE WORK

Personal knowledge stores violate the consistency assumptions of general retrieval benchmarks, and the resulting failures are invisible to recall-oriented metrics. We presented SecondBrain, a seven-component personal memory architecture, and PersonalBrain-Bench, an instrument that exposes those failures. Across four pipelines, Hit@5 saturates while MRR rises from 0.833 to 1.000, the rate of leading with superseded content falls from 0.364 to 0.000, and both post-forgetting leakage and duplicate context are eliminated — while every baseline continues to return explicitly unlearned content. Future work: an independent per-component ablation; an LLM-assisted extractor to raise contradiction recall while preserving precision; a blind human study comparing generated answers across systems; growth of the benchmark toward real-world personal corpora with appropriate privacy safeguards; and a learned, rather than fixed, weighting of the scoring terms.

ACKNOWLEDGMENT

The authors thank Dr. P S Anu Rakhi, Assistant Professor, School of Computing, for supervision and guidance, and acknowledges the open-source projects on which this system is built, in particular FastAPI, SQLAlchemy, sentence-transformers and openWakeWord.

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Artefact. All benchmark results in this paper were produced by executing the system described. Machine-readable output, including per-question retrieval traces, is emitted as `benchmark_results.json`. Benchmark generated 2026-09-22. Paper compiled 2026-09-22.